SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA51
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: Total Marks: 50 Annexure A

Code of Conduct:

I Daniel Jesuraji (192565085)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Annexure A
Analytical Problem Solving

1. RSA Encryption Analysis
Given p=11, q=13

a. Compute n ϕ(n)\
b. Choose e=7 and find the private key d
c. Encrypt the message M=9
d. Decrypt the ciphertext and verify the result
e. Analyze how changing e affects security

Given:

- $p = 11$
- $q = 13$
- $e = 7$
- Message $M = 9$

a) Compute n and ϕ(n)

$$n = p \times q = 11 \times 13 = 143$$

$$\phi(n) = (p-1)(q-1) = 10 \times 12 = 120$$

Answer:

- **n = 143**
- **ϕ(n) = 120**

b) Find private key d

Need:

$$d \times e \equiv 1 \pmod{120}$$
$$7d \equiv 1 \pmod{120}$$

Since

$$7 \times 103 = 721$$
$$721 \mod 120 = 1$$

Therefore,

$$d = 103$$

Private Key = (103,143)

c) Encrypt M = 9

Formula:

$$C = M^e \mod n$$
$$C = 9^7 \mod 143$$

Using modular arithmetic,

$$9^7 = 4782969$$
$$4782969 \mod 143 = 48$$

Ciphertext = 48

d) Decrypt

Formula:

$$M = C^d \mod n$$
$$M = 48^{103} \mod 143$$

Result:

$$M = 9$$

Original message recovered successfully.

e) Effect of changing e

- Small **e** → Faster encryption but may reduce security if used improperly.
- Large **e** → Slower encryption but generally more secure.
- Common secure value:

$$e = 65537$$

because it provides good security and efficient computation.

2. Block Cipher Mode Evaluation
Given plaintext blocks:
P=[1010,1010,1111,1010], Key = K=1100

a. Encrypt using ECB and CBC modes (Assume IV = 0000)
b. Compare the ciphertext outputs
c. Analyze which mode hides patterns better and explain why ?

Given

Plaintext blocks:
P1 = 1010
P2 = 1010
P3 = 1111
P4 = 1010
Key:
K = 1100
IV:
0000
Assume encryption uses **XOR**.

a) ECB Mode

Encryption:
$C = P \oplus K$

Block	Calculation	Ciphertext
1010	$1010 \oplus 1100$	0110
1010	$1010 \oplus 1100$	0110
1111	$1111 \oplus 1100$	0011
1010	$1010 \oplus 1100$	0110

ECB Output
0110 0110 0011 0110

CBC Mode
Block 1
$1010 \oplus 0000 = 1010$

$1010 \oplus 1100 = 0110$
C1=0110

Block 2
$1010 \oplus 0110 = 1100$

$1100 \oplus 1100 = 0000$
C2=0000

Block 3
$1111 \oplus 0000 = 1111$

$1111 \oplus 1100 = 0011$
C3=0011

Block 4
$1010 \oplus 0011 = 1001$

$1001 \oplus 1100 = 0101$
C4=0101
CBC Output
0110 0000 0011 0101

b) Compare ciphertext
ECB
0110 0110 0011 0110
CBC
0110 0000 0011 0101
Repeated plaintext produces repeated ciphertext only in ECB.

c) Analysis
ECB:
- Same plaintext → Same ciphertext
- Reveals data patterns
- Less secure

CBC:
- Uses previous ciphertext
- Same plaintext gives different ciphertext
- Hides patterns effectively
- More secure

Answer: CBC mode hides patterns better.

3. DES vs 3-DES vs Blowfish Performance Study
Given:

a. DES key size = 56 bits, time = 2 ms/block
b. 3-DES key size = 168 bits, time = 6 ms/block
c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

i. Calculate total encryption time for each algorithm
ii. Analyze trade-offs between security and performance
iii. Recommend the best algorithm for secure real-time communication

Given:

Algorithm	Time/block
DES	2 ms
3-DES	6 ms
Blowfish	1 ms

Blocks = **1000**

i) Total encryption time
DES:

$$1000 \times 2 = 2000ms$$

= **2 seconds**

3-DES:

$$1000 \times 6 = 6000ms$$

= **6 seconds**

Blowfish

$$1000 \times 1 = 1000ms$$

= **1 second**

ii) Trade-offs

Algorithm	Security		Performance
DES	Low		Medium
3-DES	Very High		Slow
Blowfish	High		Fast

iii) Recommendation

Blowfish

Reason:

- Fastest
- Strong security
- Suitable for real-time communication

4. Diffie-Hellman Key Exchange Analysis

Given:

a. p=23, g=5
b. User A private key = 6
c. User B private key = 15
d. Compute public keys for A and B
e. Compute the shared secret key
f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
g. Suggest a secure method to prevent the attack

Given

p = 23

g = 5

A private = 6

B private = 15

a) Public keys

User A

$$A = 5^6 \bmod 23$$
$$5^6 = 15625$$
$$15625 \bmod 23 = 8$$

A = 8

User B

$$B = 5^{15} \bmod 23$$

Result:

$$B = 19$$

B = 19

b) Shared Secret

A computes

$$19^6 \bmod 23 = 2$$

B computes

$$8^{15} \mod 23 = 2$$

Shared Secret = 2

c) Man-in-the-Middle Attack

Attacker intercepts public keys.
Instead of sending:
A → B : 8
Attacker sends
10
Instead of sending
B → A : 19
Attacker sends
7
Both users unknowingly establish keys with the attacker instead of each other.

d) Prevention

Use:

- Digital Signatures
- Certificates (PKI)
- Authenticated Diffie–Hellman
- TLS/SSL

These authenticate the public keys and prevent MITM attacks.

5. ECC vs RSA Comparison

Given:

a. RSA key size = 1024 bits
b. ECC key size = 160 bits (equivalent security)
c. Device processing capacity = limited (IoT device)
d. Analyze computational cost for both systems
e. Estimate which consumes less power and memory
f. Justify which cryptosystem is better for the given scenario with reasoning

Given:

- RSA = 1024 bits
- ECC = 160 bits
- IoT device with limited resources

a) Computational Cost

RSA	ECC
Large key	Small key
More CPU operations	Fewer CPU operations
Slower	Faster

b) Power and Memory

Feature	RSA	ECC
Power	High	Low
Memory	High	Low
Execution	Slow	Fast

ECC consumes less power and memory.

c) Justification

ECC is the better choice for IoT devices because:

- Smaller key size (160 bits)
- Equivalent security to 1024-bit RSA
- Faster computation
- Lower memory usage
- Lower power consumption
- Ideal for battery-powered and resource-constrained devices

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)

Problem Understa ndi ng	20%	Clearly interprets problem, identifies all parameter s and requiremen ts accurately	Understa n ds problem with minor gaps	Partial understa ndi ng; misses key elements	Misinter pre ts or unable to understa nd problem
Method Selection	20%	Selects most appropriate method/for m ula with strong justification	Selects correct method with minor errors	Inappropri at e or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorgani ze d steps; multiple errors	No clear process
Accuracy of Calculatio n	20%	All calculatio ns accurate; correct final answer	Minor calculati on errors	Frequent errors; incorrect final answer	No valid calculati on s
Interpreta tio n of Results	15%	Clearly interprets results with units and engineeri ng relevance	Basic interpret ati on with minor omissions	Limited or unclear interpreta tio n	No interpret ati on